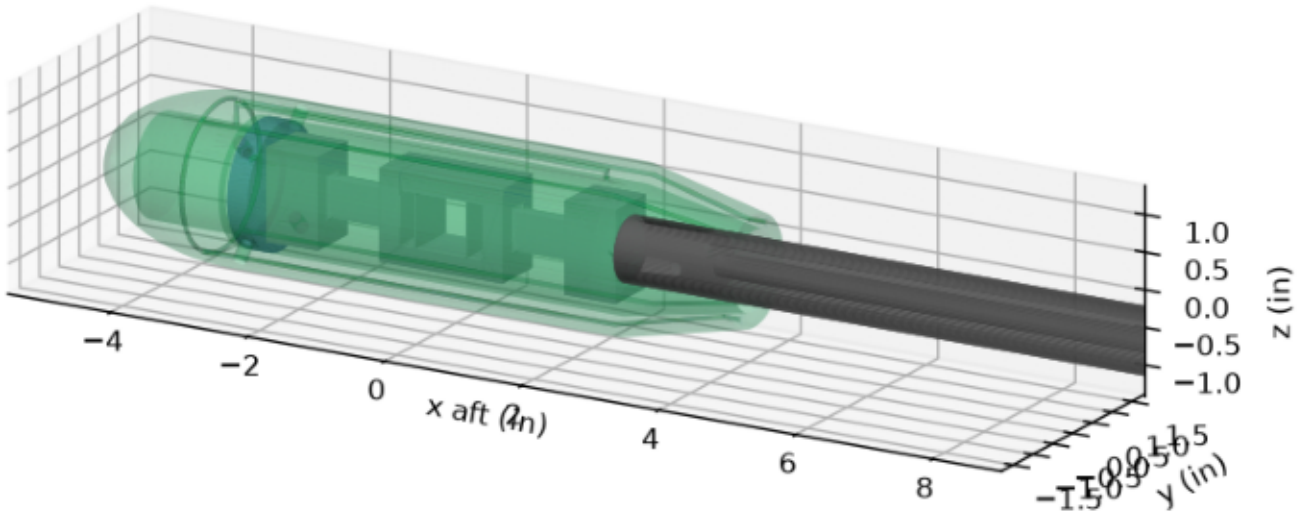


FLIGHT ASSEMBLY (CAD RENDER, NTS)

Flight assembly: instrumented 2.25 in pod (transparent), puck, 6-component balance, sting



NOTES:

1. GAUGE AND WIRE THE BALANCE (24 GAUGES, 6 BRIDGES) BEFORE ASSEMBLY; SEE THE GAUGE LAYOUT DRAWING.
2. PUCK TO BALANCE FRONT FACE: 2 DOWELS + 2 SCREWS. STING ON THE SPIGOT: CROSS PIN + 2 SET SCREWS.
3. WIRES RUN AFT THROUGH THE BALANCE AND THE STING BORE TO J0 AT THE ROLL HOUSING.
4. THE MODEL POD IS 3D PRINTED FROM THE STEP FILE (NO MACHINING DRAWING).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

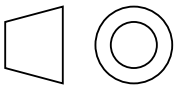
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-001	Balance, 6-component (gauged)	1	7075-T651
2	WT-6C-002	Puck, model interface	1	6061-T6
3	WT-6C-003	Sting	1	4140 28-32 HRC
4	POD-2.25	Pod, reference model (STEP)	1	3D printed
5	-	Strain gauge 350 ohm, CEA-13 class	24	Vishay / bought
6	-	Dowel pin 1/8 x 1/2	2	bought
7	-	SHCS #10-32 x 5/8 (puck)	2	bought
8	-	Cross pin 3/16 dowel (sting)	1	bought
9	-	Set screw 1/4-28 cup point	2	bought
10	-	SHCS #8-32 x 3/8 (pod)	4	bought
11	-	Wire 32 AWG PTFE + J0 DC-37	1	see Electronics

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, FLIGHT (BALANCE + STING + MODEL)								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-100		A	B	NTS		-		1 OF 35